

SEMINAR III · WEEK 4 · FOUNDATIONS

Fractions & Decimals

Two ways to write the exact same amount. Once you can move between them, a lot of math gets easier.

SESSION A

Fractions: Meaning to Operations

What a fraction means · comparing them · adding, multiplying, dividing

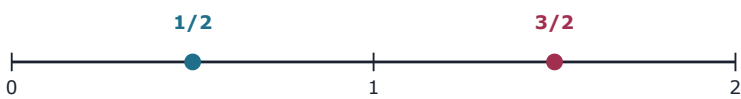
WHAT IS A FRACTION?

Look at $\frac{3}{4}$.

The bottom number, 4, is the **denominator** — how many equal parts one whole has been divided into. The top number, 3, is the **numerator** — how many of those parts you have. A fraction also means division: $\frac{3}{4}$ also means $3 \div 4$, which is 0.75.

FRACTIONS ARE POINTS, NOT JUST PIECES

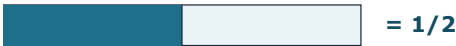
Fractions don't have to be less than 1.



One half sits exactly halfway between 0 and 1. Three halves is greater than 1 — it sits halfway between 1 and 2.

EQUIVALENT FRACTIONS

Same amount, different name.



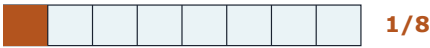
Multiply the numerator and denominator by the same nonzero number and the value doesn't change — the shaded amount is identical in all three bars.

SIMPLIFYING FRACTIONS

12/18 simplifies to 2/3.

Both numbers divide evenly by 6: $12 \div 6 = 2$, $18 \div 6 = 3$. The way it's written changed. The quantity did not.

WHICH FRACTION IS BIGGER?



More pieces means smaller pieces — 1/8 is **less than** 1/4, even though 8 is a bigger number than 4. When denominators match, just compare numerators. When they don't, convert to decimals or use a benchmark like 1/2.

ADDING FRACTIONS

Same-size pieces first.

$\frac{1}{2} + \frac{1}{3} \neq \frac{2}{5}$. The pieces have to be rewritten with a common denominator before you can add them.

Like denominators: add numerators directly, $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$.

Unlike denominators: find a common denominator first,
 $\frac{1}{2} + \frac{1}{4} \rightarrow \frac{2}{4} + \frac{1}{4} = \frac{3}{4}$.

MULTIPLYING & DIVIDING FRACTIONS

Multiplying means "part of a part."

$\frac{1}{2} \times \frac{3}{4}$ means half of three fourths — multiply straight across: $1 \times 3 = 3$, $2 \times 4 = 8$, so **$\frac{3}{8}$** . No common denominator needed; that rule is only for addition and subtraction.

Dividing asks a different question: $\frac{3}{4} \div \frac{1}{2}$ asks how many halves fit into $\frac{3}{4}$ — the answer is $1\frac{1}{2}$. Shortcut: keep the first fraction, flip the second, multiply.

BEFORE WE MOVE ON

1. Simplify a fraction
2. Compare two fractions
3. Explain why two fractions are equivalent

SESSION B

Decimals & Choosing a Representation

Decimal place value and operations · fractions and decimals as the same number, different form

UNDERSTANDING DECIMALS

Every digit after the decimal point has a specific place value.

In 4.583, the 5 is tenths, the 8 is hundredths, the 3 is thousandths — that 8 represents 0.08, not just "the number 8." Adding zeros doesn't change the value: 0.5 = 0.50 = 0.500.

Is 0.58 greater than 0.6, because 58 looks bigger than 6?
No — rewrite with matching digits: 0.60 vs 0.58. Place value decides, not digit count.

DECIMAL OPERATIONS

Adding or subtracting: line up the decimal points so you're combining digits with the same place value.

Multiplying, especially with a calculator: always estimate first. 4.9×3.1 should land close to $5 \times 3 = 15$. If your calculator shows 151.9, something went wrong — probably a decimal point in the wrong place.

FRACTIONS AND DECIMALS ARE CONNECTED

Since a fraction means division, convert a fraction to a decimal by dividing: $3/4 = 3 \div 4 = 0.75$. Going the other way, a decimal becomes a fraction using its place value: $0.6 = \text{six tenths} = 6/10 = 3/5$.

$1/2 = 0.5$

$1/4 = 0.25$

$3/4 = 0.75$

$1/5 = 0.2$

$2/5 = 0.4$

$1/10 = 0.1$

These benchmarks come up constantly — they're worth knowing cold. We'll use them again next week for percent.

WHICH FORM SHOULD I USE?

Money is almost always a decimal: \$2.75, not \$2³/₄. Recipes usually use fractions: 3/4 cup, not 0.75 cup. When comparing two values, decimals can make it easier: converting 7/8 to 0.875 makes it immediately clear that 7/8 > 0.82.

BEFORE WE MOVE ON

1. Convert one fraction to a decimal
2. Convert one decimal to a fraction
3. Explain which form is more useful in a given context

APPLY, ANALYZE, RETRY

Mixed Practice & Week 4 Check

TODAY'S PLAN

- Mixed retrieval: fraction meaning, comparison, operations, decimal operations, conversions, reasonableness
- Classify anything you miss by error type, then retry after feedback
- Week 4 check, then reflection: one strength, one developing skill, one error pattern, one strategy for next week

TRANSITION TO WEEK 5

Percent

Fraction $\leftrightarrow$ Decimal $\leftrightarrow$ Percent, e.g. $1/2 = 0.5 = 50\%$. Applied to discounts, tips, tax, wages, and grades.